Auditing the Winner: An Upper Bound on What Critic Disagreement Adds Beyond the Selector on SWE-Agent Trajectories

Anonymous submission
Paper under double-blind review

***Abstract*—A coding agent tries a task several times and a scoring model picks its best attempt. We checked those picks against the projects' real test suites on two SWE benchmarks: 35% and 45% of the committed trajectories were silent failures — the agent said it was finished, the scorer agreed, and the tests failed. The safeguard we tested, pre-registered throughout, is disagreement among a disjoint panel of critics. On the population the selector actually commits, against the selector's own score, with executed gold, disagreement ranks failures at 0.508 and 0.565. Its pooled incremental estimates range from -0.030 to $+0.013$ AUROC; every interval includes zero, and the loosest upper bound is $+0.12$ over 47 failures and 46 repositories. Because that bound exceeds the $+0.05$ we declared actionable, this is a failure to detect gain, not evidence of absence. The critics are consistent and individually informative; measuring how much they differ discards that information. Promising pilot results vanished on three times as many repositories, seven instrument failures would each have produced a convincing false finding, and on a stronger agent no critic cleared the registered selector bar, so that cell is descriptive only. A single strong judge's own score, added after the fact, does raise detection, by $+0.17$ AUROC pooled: the critics' judgments carry; their disagreement does not.**

I. Introduction

Best-of-n selection under a learned critic is a default in agent scaffolds: generate several trajectories, score each with a reward model, commit the argmax. Three claims about this are easy to conflate, and this paper keeps them apart. The *theory* is that argmax selection on a noisy score induces the optimizer's curse [1]: the top-scored candidate is disproportionately the one whose flaws the critic cannot see, so the score carries less information about the committed candidate than about the pool. What we *observe* is one instance of that theory measured directly: on Verified the selecting critic ranks failures at 0.81 among candidates it passed over and 0.61 among those it committed, on the same instances. What we do *not* show is that selection raised the failure rate of the committed candidate — with the selector we used, it did not (§II). A committed trajectory that fails while carrying a high score is a *silent failure*; they are a third to a half of what this agent committed, whether or not selection put them there.

The obvious safeguard is a second opinion. If a panel of critics disjoint from the selector disagrees sharply about a committed trajectory, that disagreement might carry information the selector has exhausted. That correlated critics share blind spots is not our idea: the correlated-judge literature predicts it [2–6], and best-of-n reward hacking is well characterised [7–9]. What had not been done is the measurement: put a disjoint panel on the population a critic actually committed, with executed gold, structural baselines and pre-registered bars, and ask what it adds over the selector. Our contributions:

- **A selection-conditional validation protocol for audit signals** — evaluate on the committed population, compare against the selector's own score, use executed gold, and cluster on the real dependence unit — with its first two-benchmark instantiation on SWE-agent trajectories. This is a reusable harness test for an auditor, not a new auditor. Candidate-pool discrimination does not measure residual auditing value on the selected population [10, 11].
- **A replicated failure to detect incremental gain, with an estimator-robust bound**: the pooled estimates range from -0.030 to $+0.013$ and the loosest upper bound is $+0.12$ over 47 positives and 46 repositories, above the registered $+0.05$ actionable margin.
- **Diagnostics**: test–retest stability of long-window reward-model scores, the repository-clustering ceiling on effective n, and seven instrument failures that each would have produced a false positive (§III-C).

"Critic" means an off-the-shelf reward model or LLM judge used without task-specific training. Trained trajectory verifiers [12–14] are a different class.

a) Setup: The unit is one committed trajectory. Gold $g \in \{0, 1\}$ is trajectory-level success under the official `swebench` 5.x harness; the positive class is failure (g=0). The selecting critic c_{sel} scores each of n=4 full candidates and commits the argmax; the audit panel $\{c_1, \ldots, c_m\}$ satisfies $c_{\text{sel}} \notin \{c_i\}$ at checkpoint level, and u is the sample standard deviation of their rank-normalised scores, with rank norms fit on training repositories only. The population is committed trajectories whose agent exited with a submission and a non-empty patch. In numbers: 41 Verified instances gave 34 committed rows (7 excluded for an empty patch or an ungradable run); 114 Multilingual instances gave 77 (30 empty patches, 7 harness errors). M1 $= \text{AUROC}(u)$, M2 $= \text{AUROC}(-c)$, M3 $=$ M1 $-$ M2 with a repository-clustered bootstrap. Population, positive class, metrics, predictions and a stopping rule were committed to a version-controlled pre-registration before any gold existed; each later analysis was added as a dated amendment before its data existed.

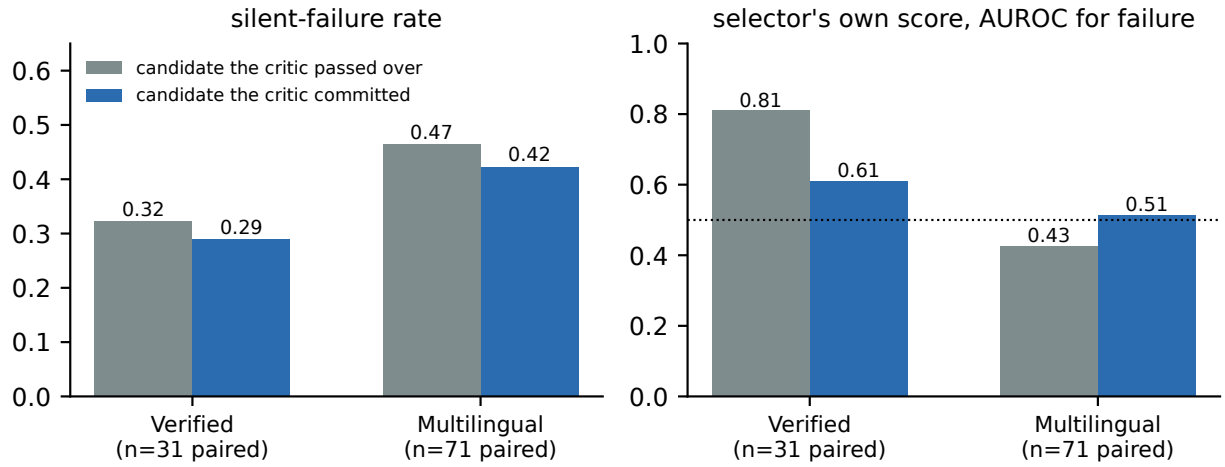


Fig. 1. **Picking did not make things worse — but it did use up the picker's own judgment.** One candidate the critic passed over, graded per instance, paired with the one it committed.

II. Silent failures are common; increased failure under selection was not detected

Of 164 Verified candidates, 138 exited having submitted; selection by `Skywork-Reward-V2-Qwen3-4B` committed 34 usable trajectories, of which the harness resolved 22 and failed 12: **35.3% [21.5, 52.1] silent failures**. On Multilingual the same pipeline gave 77 usable rows and 35 failures (45.5%). Selection was real (the chosen index spread across all four slots) and $\mathrm{corr}(c, u) = 0.11$. One incidental measurement bounds what *any* auditor on this benchmark can be checked against: the median analysed instance is guarded by exactly one `FAIL_TO_PASS` test, and 84 of 111 analysed rows are guarded by exactly one (22 of 34 on Verified, 62 of 77 on Multilingual). "Resolved" is usually a single test's verdict, so the label our AUROCs are computed against is thinner than the word suggests.

Whether selection *concentrated* failure is a separate question a committed-only sample cannot answer. A pre-registered control graded one unselected candidate per instance on both benchmarks (31 and 71 paired rows). Figure 1 shows both arms. **We did not detect increased failure under selection**: -3 points on Verified $[-21, 0]$ and -4 on Multilingual $[-11, +3]$. The prediction that selection concentrates failure (**F-P1**) failed on both; the Multilingual interval still admits three points more concentration, so this is a failure to detect rather than a demonstrated absence, and we claim a failure *rate*, not concentration. A second registered control (**F-P3**) required disagreement on the random arm to be reported: it is 0.367 on Verified and 0.51 on Multilingual, so the range-restriction reading — disagreement working on an unselected pool while failing on the committed one — is not available either. The control does measure the selection-conditional gap: on Verified the selector ranks failures among candidates it did not pick at 0.81 [0.74, 1.00] and among those it committed at 0.61 on the same instances. On Multilingual the critic has no signal in either arm (0.43, 0.51), so there was nothing to remove.

a) Disagreement did not reliably detect them: The primary analysis — four chat-preference reward models auditing a 4,096-token prefix — gave $\mathrm{AUROC}(u) = 0.508$ against the selector's own 0. The four are `GRM-gemma2-2B`, `GRM-Llama3.2-3B`, `Skywork-Reward-V2-Llama-3.2-1B` and `Skywork-Reward-V2-Qwen3-1.7B`; the selector is `Skywork-Reward-V2-Qwen3-4B`. In plain terms: u is how much the panel disagrees about one trajectory (the standard deviation of their rank-normalised scores); $-c$ is the selector's own score with the sign flipped, so higher means more likely to fail; LORO is leave-one-repository-out, so every score is predicted for a repository the model never saw.621. Every extension was declared before its data existed and asked whether some limitation of the primary was hiding a signal: the final diff plus the last 3k tokens, the patch in view, a true 16k context, re-selection by the proposal's named 8B critic and by a code-trained model, `AceCodeRM-7B` and its 32B sibling, and a confirmatory panel of three `gpt-5`-family judges. Panel u never left 0.439–0.686 (Fig. 2); the two panel values quoted here, 0.494 with the patch in view and 0.439 at 16k, are the registered four-member panel, not the six models scored individually. Individual judges did better: `gpt-5` reached 0.720 and `gpt-5-mini` 0.752, while their disagreement was 0.528. `gpt-5-mini` is also the agent, so its 0.752 is a self-evaluation; `gpt-5` is not, and clears the same bar at 0.720. The registered reading (**A4c-P1**) would therefore narrow the claim to *reward-model* critics; we decline that narrowing because the judge saw the diff and the reward models did not, and label the decision post hoc. *Every registered prediction that panel disagreement carries signal beyond the selector failed.* One registered bar was cleared and we disclose it: the primary hypothesis was the bare inequality $\mathrm{AUROC}(u) > 0.5$, and u is 0.508 and 0.565,

so it passes as written. We do not treat it as a detector — the bar carries no minimum effect size, both intervals include chance, and the paired comparisons are -0.114 and $+0.005$. Registering a threshold with no margin was our design error; overruling it is post hoc and labelled so. Two registered own-score predictions also fail: the Goodhart prediction (**P2**) required $\mathrm{AUROC}(-c) \leq 0.5$ and fails at 0.621, while the code-trained critic prediction (**A2-P1**) required AUROC above 0.60 and fails at 0.405. One prediction, on that critic's prefix-versus-full-length correlation, is undecidable: it fell back to 4k on 31 of 34 rows. Figure 2 plots all thirteen. Panel disagreement spans 0.439 to 0.686 across every variation of input length, critic size, critic identity, training signal, critic class and selection regime, against a selecting critic at 0.621 and a structural baseline at 0.665. The paired gain M3 is negative in nine of thirteen rows and no interval excludes zero; the widest upper bound anywhere in the sweep is $+0.63$, on the code-trained critic's 16k re-selection over eleven positives. Figure 2 plots every row; the thirteen configurations are enumerated above.

One configuration crossed its bar. Amendment 3 pruned tool outputs and scored overlapping 3,500-token windows, taking the panel's maximum disagreement: $\mathrm{AUROC}(u_{\max}) = 0.686$ against a pre-registered bar of 0.665, with a free permutation $p = 0.04$ that rises to $p = 0.21$ within repository and a leave-one-instance-out range of 0.657–0.731 that straddles the bar.

b) A ruler beats the critics: Two integers read off the diff and the transcript, with no model involved, rank silent failures better than every reward model at 4k and every single-window disagreement statistic at any length: lines added, 0.665, and agent step count, 0.657 (Fig. 2). We propose lines-added as the bar any audit statistic must clear before it earns a confirmatory run. It is a screening baseline and not a deployable guardrail, since an agent optimising against it would simply write shorter patches; §III shows it does not survive the replication either, which is the point.

III. Replication on 37 repositories

The binding limitation of the pilot is nine repositories. Amendment 4e, pre-registered before any trajectory existed, re-ran the instrument on SWE-bench Multilingual: 41 repositories in nine languages (37 survive into the analysed population), 114 instances, 456 trajectories, the same selector and panel, agent sandboxes and gold on Modal, with three reference-patch / empty-patch oracle pairs passed before any model patch was graded.

R-P1: panel u 0.565 [0.43, 0.71] against the critic's 0.560; paired difference $+0.005$ [-0.19, $+0.22$]; $p = 0.85$. Registered rule met ($\mathrm{AUROC} < 0.60$, CI includes 0.50), but upper bound 0.71 — we fail to reject, not establish equivalence (same reading as Amendment 7a). **R-P2** (the windowed positive confirms): $u_{\max}$ 0.541 [0.40, 0.68], within-repository $p = 0.53$, leave-one-out 0.528–0.557. *Failed.* **R-P3** is vacuous: $u_{\max}$ exceeds the lines-added baseline only because that baseline itself fell from 0.665 to 0.508 [0.35, 0.65]. Patch size predicted silent failure on Verified and does not here. The best reward model at 16k is 0.577 (the 1B model that scored 0.712 is at 0.485); the named 8B critic 0.52; the code-trained model 0.356; LLM judges 0.58–0.66 with disagreement 0.514.

A. The registered endpoint

The proposal's confirmatory quantity was the held-out $\Delta\mathrm{AUROC}$ of M_{c+u} over M_c. The 70/30 repository split never held ten test positives, so it was never eligible on the pilot alone; with 47 positives across both benchmarks it can be computed by leave-one-repository-out, a declared substitution of the cross-validation scheme, not of the estimand. Because a two-parameter model refit on forty-five clusters out of forty-six is still a fragile thing to hang a bound on, we compute it four ways (Table I). They disagree in sign. They agree on the conclusion: every interval includes zero, and among the three estimators that use every row the loosest pooled upper bound is $+0.12$ ($+0.25$ on the pilot alone). The fixed-weight variant reproduces the raw $\mathrm{AUROC}(-c)$ almost exactly (0.559 vs. 0.560) while the logistic variants report M_c at 0.48–0.50: pooling out-of-fold probabilities across folds with different intercepts scrambles the global ranking, depressing both arms and largely cancelling in the difference. With rank metrics and few clusters this can make a real signal appear to vanish; an unfitted combination is the cheap check.

B. A capable selector, and newer judges

Amendment 5 re-scored every candidate in both pools with `gpt-5` and committed its argmax (620 candidates, $13, 59 new patches graded). A defect found in review must be stated first: the analysis filtered its population by the *4B*-committed candidate's agent-declared-complete flag rather than the `gpt-5`-committed one, which on Multilingual dropped 16 eligible instances. Verified is unaffected; the corrected Multilingual set is 101 rows, 44 failures and 40 repositories (not the 77-row primary set: a different selector commits different trajectories), and the numbers below are the corrected ones.

S-P1 passes on both registered conditions: pool $\mathrm{AUROC}(-c)$ of 0.712 and 0.706, and a committed failure rate below the random arm's, paired within instance, at -0.065 [-0.20, 0.00] and -0.084 [-0.14, -0.03], with no pair where the committed trajectory failed and the random one passed. S-P2 fails on both: the gap is -0.080 on Verified and 0.035 on Multilingual against a registered bar of 0.10, so we withdraw the claim that selection measurably consumes a strong selector's signal and keep only the 4B observation on Verified (0.81 to 0.61). S-P3 fails on its first condition, $\mathrm{AUROC}(u) = 0.591$ against a bar of 0.60, and the endpoint is small under every estimator: $+0.012$, -0.024, and -0.018 [-0.11, $+0.07$] unfitted (signs still disagree; none is large and none has a degenerate baseline). The defect also explains the study's most alarming number: the registered logistic's $+0.375$ [$+0.21$, $+0.52$], with a baseline collapsed to 0.150, becomes $+0.012$ with a baseline of 0.575. That cell is *consistent with* the null and does not establish it, since the

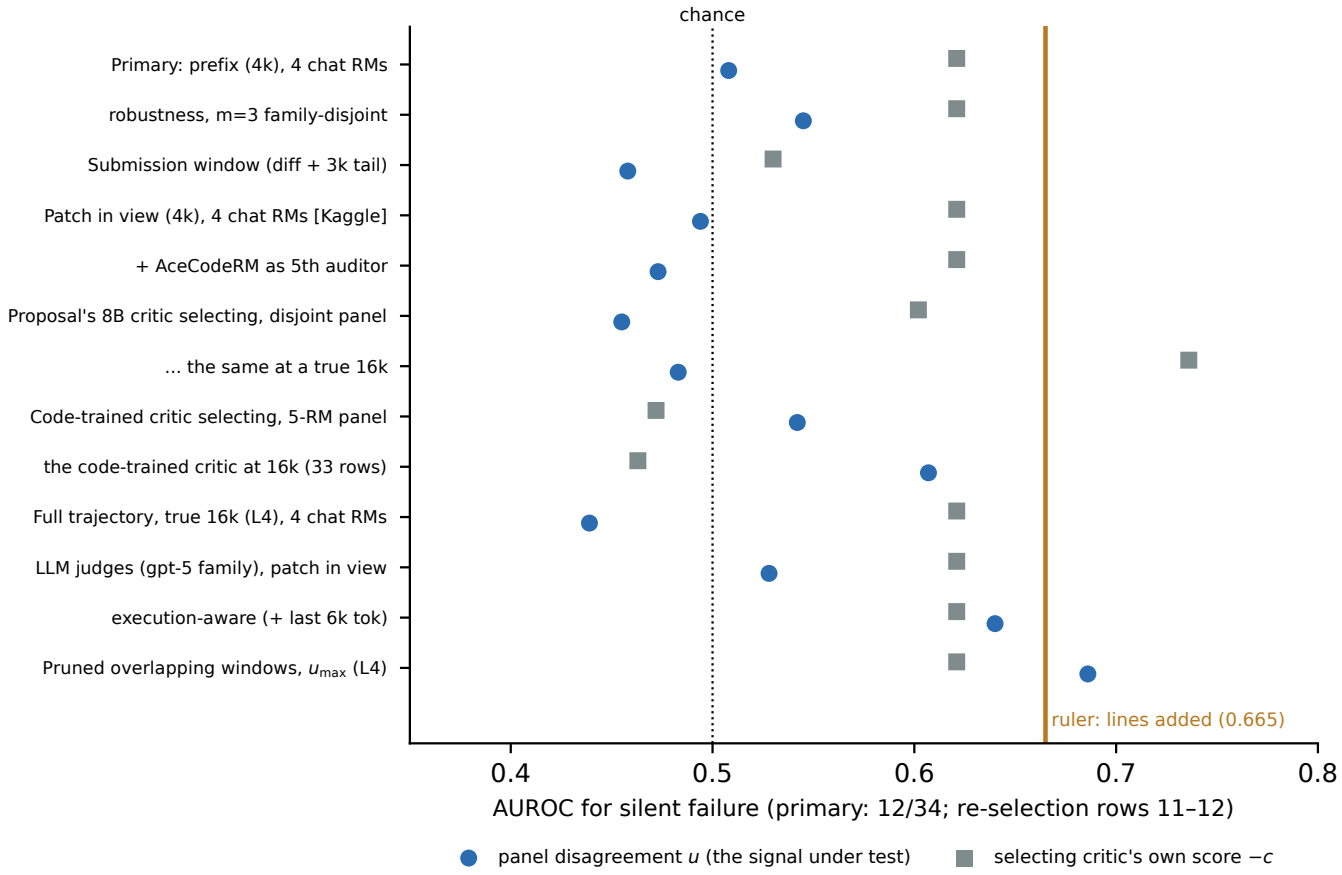


Fig. 2. **Every way we tried to make disagreement work, on the pilot.** Each row is one configuration; 0.5 is a coin flip. Circles are panel disagreement, squares the selecting critic's own score. The gold line is lines added to the patch: a baseline that needs no model at all. Exactly one circle crosses it: pruned-window $u_{\max} = 0.686$, which failed to replicate. Exactly one square does: the 8B critic's own score at a true 16k, 0.736.

upper bound of $+0.073$ exceeds the $+0.05$ we declared actionable.

Newer judges sharpen the dissociation rather than resolving it. `gpt-6-sol` reaches 0.776 [0.68, 0.87] on the replication and 0.826 on the pilot, the best single detector in the study, and both `gpt-6` judges beat every `gpt-5` judge on both benchmarks, while the newer pair's disagreement falls to 0.41 against the weaker trio's 0.51 and a five-judge panel gives 0.44 [0.31, 0.58]. Better critics, no better disagreement. Exploratory, one vendor, and a two-member spread is a weaker statistic than a four-member one.

a) Post hoc: a single strong judge does add to the selector: Two reviewers asked the obvious next question, so we answered it after the fact, under the same protocol: the committed population, the selector's score as the baseline, leave-one-repository-out, a repository-clustered bootstrap. Adding `gpt-6-sol`'s own score to the selector's score raises failure detection by $+0.18$ [$+0.10$, $+0.28$] on Multilingual, $+0.14$ [-0.02, $+0.28$] on Verified and $+0.17$ [$+0.10$, $+0.25$] pooled (fixed unit weights; the logistic estimator agrees). This was not pre-registered, the judge saw the patch, and the pooled interval carries the small-sample caveats above. But it is the practical reading of the dissociation: a critic's judgment carries beyond the selector; its disagreement with other critics does not.

b) Mechanism: stable critics, uninformative disagreement: Re-scoring an identical window after a content-preserving reformatting, every critic reproduces its ranking at $r = 0.98$–0.99; after a 128-token shift, at 0.86–0.95; the per-window panel u at 0.91 and 0.73. Adjacent windows with 25% overlap correlate at 0.20–0.56, first against last at -0.39 to 0.47. These critics are stable instruments responding to what is in front of them: the prefix score judges the problem, the patch score judges the patch, and they differ because the inputs do. An earlier draft read that difference as critic instability; this test withdraws that claim. The failure is in the statistic. A panel's disagreement is large when its members' rank-normalised scores spread out. Post hoc, critic location accounts for under 1% of rank-scale variation; 29–44% is shared trajectory level and 56–71% is residual critic-by-trajectory variation, inseparable from cell-level score noise. That residual supplies the within-trajectory spread disagreement is computed from.

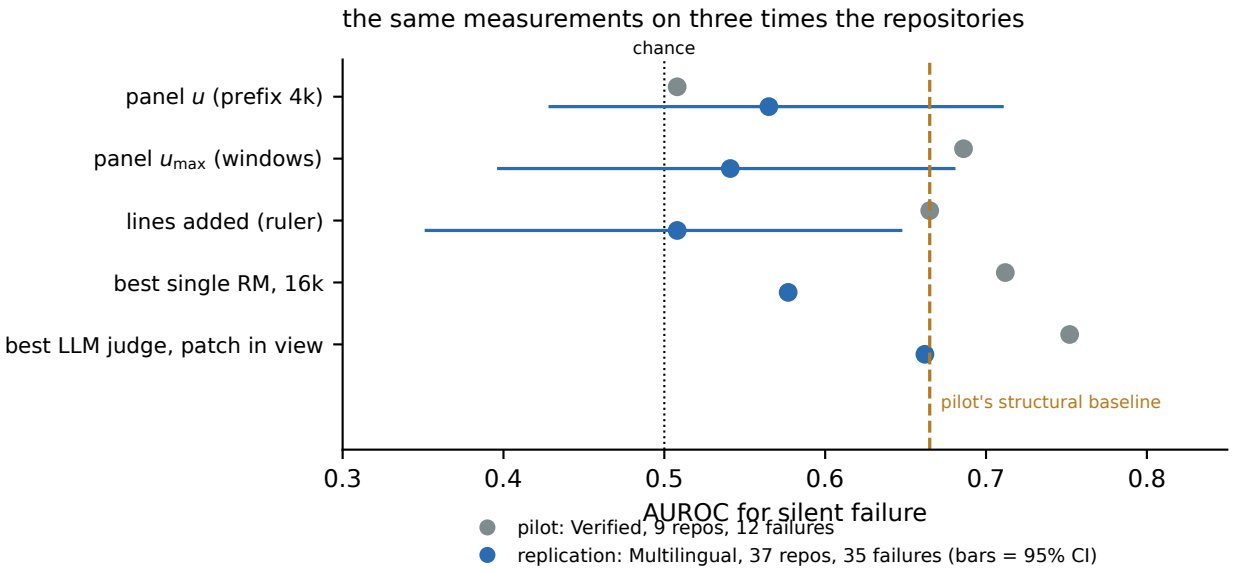


Fig. 3. **The pilot's disagreement and reward-model positives did not travel; single LLM judges remain above chance.**

Estimator	Population	base	$+u$	ΔAUROC	95% CI
Logistic, leave-one-repo-out (pre-registered)	Pooled	0.484	0.454	-0.030	$[-0.10, +0.04]$
Logistic, 5 clustered folds	Pooled	0.504	0.491	-0.014	$[-0.09, +0.07]$
AUROC within fold, size-weighted	Pooled	—	—	-0.119	$[-0.31, +0.28]$
Fixed unit weights, no fitting	**Pooled**	**0.559**	**0.571**	$\mathbf{+0.013}$	$[\mathbf{-0.07}, \mathbf{+0.12}]$

TABLE I

THE REGISTERED CONFIRMATORY ENDPOINT UNDER FOUR ESTIMATORS, REPOSITORY-CLUSTERED BOOTSTRAP; 111 ROWS AND 47 POSITIVES, OF WHICH MULTILINGUAL SUPPLIES 35. THE LAST THREE ARE POST-HOC ROBUSTNESS CHECKS OF THE FIRST, AND POOLING THE TWO BENCHMARKS WAS ITSELF ADDED AFTER REGISTRATION; THE PER-BENCHMARK LORO ROWS ARE WHAT WAS REGISTERED. THE WITHIN-FOLD ESTIMATOR DROPS REPOSITORIES WHERE NO AUROC IS DEFINED AND COVERS 53 OF 111 ROWS. **PER BENCHMARK THE UNFITTED ROWS DISAGREE IN SIGN** — VERIFIED -0.030 $[-0.17, +0.25]$, MULTILINGUAL $+0.048$ $[-0.07, +0.18]$ — AND BOTH PER-BENCHMARK UPPER BOUNDS ARE LOOSER THAN THE POOLED ONE; THE LEAVE-ONE-REPOSITORY-OUT ROWS ARE -0.076 $[-0.16, +0.08]$ AND -0.022 $[-0.13, +0.08]$. POOLING IS WHAT 47 POSITIVES SUPPORT, NOT THE LOOSEST STATEMENT AVAILABLE.

C. *The instrument, and seven ways it lied to us*

Each produced a plausible result that was wrong, and each was caught by an oracle control, a regression test, or a registered audit before a number was trusted. (1) A sandbox flag hid `numpy` from the gold harness, scoring every trajectory a failure; caught because reference solutions run through the harness first. (2) An LLM-as-judge selecting critic emitted three distinct values over 640 candidates, so best-of-16 silently became first-draw and 39 of 40 committed trajectories matched the $n{=}4$ run; caught by the chosen-index histogram. (3) Harness image pulls failed on an architecture manifest: 41/41 failures in 499 seconds; caught by the harness's own error count. (4) Our power function reported $\approx$0.40 power under the true null against a correct 0.010; caught by simulating the null. (5) Repository grouping truncated hyphenated names, so held-out repositories were not held out. (6) Rank normalisation was fit on all rows, leaking test rows into u; every u here is post-fix, verified against raw panel scores (max absolute difference 0.0). (7) The selector-axis analysis filtered by the *previous* selector's committed candidate, dropping 16 eligible instances and manufacturing $+0.375$ from a collapsed baseline; on the correct population it is $+0.012$ (§III-B). The rule we adopted: a rate of exactly 0% or 100%, or a statistic that cannot move, is an instrument, not a result.

IV. LIMITATIONS

- **One disagreement statistic; one kind of failure.** We measured disagreement as the spread of rank-normalised scores. Five other functions of the same panel scores did no better (best 0.601), but richer notions of disagreement, such as which critics disagree and about what, were not tried. Failures that tests cannot see, such as unauthorised actions or departures from the user's instructions, were out of scope; disagreement may be more informative there.
- **A capable open-weight agent: the axis ran, and no critic qualified as its selector.** `DeepSeek-V4.1-Flash` ran on the same 41 Verified instances, scaffold, 40-step limit and harness gold, against gates registered before the run. It is the *more* capable agent — 30 of 41 resolved, an unconditional failure rate of 0.268 against `gpt-5-mini`'s 0.486, at $0.0135 per trajectory — but submitted on 32 of 41 against an 80% bar, failing the functioning gate by one

instance; all nine misses exhausted the step budget rather than failing to parse. Raising it would break comparability with the commercial agent that passed at 41 of 41, so the 40-step axis was not launched under Amendment 10. `GLM-5.3`, run under the same 100-step, 16k-token settings for completeness, submitted on 16 of 41 and every one of those resolved, but 23 of 41 runs were cut by a $0.25 per-trajectory cost cap the amendment had not named — inert for DeepSeek at $0.015 a run, binding for GLM at $0.22 — so the budget question is unanswered for GLM and it remains disqualified on cost at any budget. Both failed a bar set by our own 40-step budget, so this is in part a finding about *this scaffold* rather than about the models: DeepSeek resolved 30 of the 32 instances where it did submit. We do not generalise it to the models. At a 100-step budget, declared post hoc, DeepSeek submitted on 39 of 41 (0.951) with none reaching the ceiling (longest 79 steps), and its failure rate of 0.098 against `gpt-5-mini`'s 0.610 at the same budget (0.556 excluding five self-inflicted test-suite crashes) clears every registered gate. The axis then ran under Amendment 13: 456 attempts over the registered 114 Multilingual instances at 100 steps, all graded (365 of 456 resolved, no harness errors). **A7-P1** passed — DeepSeek's unconditional failure rate 0.200 against 0.579 for `gpt-5-mini` at the same 100-step cap (DeepSeek made four attempts per instance, 456 in all; `gpt-5-mini` made one, 114 in all) — but the registered selector gate failed: the `gpt-5` judge's pool AUROC for detecting the agent's failures is 0.552 against a bar of 0.65, and the five reward models scored on this axis sit between 0.419 and 0.523. A stronger judge does not change this: as selectors on the same pool, `gpt-6-sol` reaches 0.619 and `gpt-6-luna` 0.569 (Amendment 14), both below the bar, and their picks fail more often than the frozen random picks (0.193 and 0.211 against 0.175); the judges return integer scores and tie at the top on 41 and 38 instances, which the readout breaks toward the first attempt — broken toward the last, `gpt-6-sol`'s picks fail at the random rate (0.175) and `gpt-6-luna`'s at 0.202, so for `gpt-6-sol` the "worse than random" reading depends on the tie rule. With no critic above the selector bar of 0.65, the committed sets hold 17 and 19 failures against a floor of 20, and the cell is reported as descriptive only (primary ΔAUROC $+0.020$ $[-0.120, +0.149]$ on the judge arm). The objection that the null is specific to weak agents is not answered by an equivalence bound; we read it as re-posed as a question about selectors, which is interpretation, not a registered result.

- **One agent; two selectors, both off the shelf.** gpt-5-mini at 40 steps throughout the main study (Amendments 11 and 13 ran at 100). The primary selector is a 4B chat reward model, so the primary selected population is that of a weak selector; §III-B re-selects with `gpt-5`. Neither is a deployed-grade trajectory verifier. The capable-agent cell was run under Amendment 13 and is descriptive only.
- **The selection-conditional regime is only partly closed.** On Multilingual the primary 4B selector had no signal in either arm, so residual value after selection is measured for that critic only on the pilot. Amendment 5 re-selects with `gpt-5`, which ranks the pool at 0.71 on both benchmarks; on the corrected population S-P2 fails (gap 0.035 against a bar of 0.10) and S-P3 fails on $\mathrm{AUROC}(u) = 0.591$ against 0.60, so the strong-selector cell is consistent with the null and does not establish it (upper bound $+0.073$). The capable-agent cell is the Amendment 13 result above: capability passed, the selector gate failed, and the disagreement endpoint is descriptive only.
- **Critic classes.** Six chat-preference RMs, two code-trained RMs, five LLM judges from one vendor across two generations. Cross-vendor judges and trained trajectory verifiers were not run; the claim is scoped to off-the-shelf critics.
- **Gold is the harness verdict**, which accepts some incorrect patches and rejects some correct ones [15–17]; failure rates are lower bounds and AUROCs attenuated. Bounding the lucky-pass direction: no passing trajectory edited a test file, 4 of 64 passes are disjoint from the reference patch, and dropping or relabelling those four moves the pooled endpoint to -0.053 and -0.024 — further from the safeguard working, not towards it.
- **Forty-seven positives.** Intervals are ± 0.25 (pilot) and ± 0.14 (replication). Roughly twenty-five analyses were run on the twelve pilot positives without multiplicity correction; the replication is the correction. The study supports failure to reject and an upper bound, not equivalence to chance.
- **Repository clustering.** Verified is 500 instances over 12 repositories (django 46%); the public ICC of resolution by repository is 0.033–0.053, an effective n of 64–95 of 500. The paired "beat the baseline by 0.10" test has power $\approx$0.3 at any n, which is why the replication changed repositories rather than instances.
- **Exploratory analyses.** The diversity-versus-error-correlation experiment was run after the confirmatory results: no decorrelated panel exists among these critics, all 35 four-member panels sitting at pairwise error correlation 0.71–0.89, and within that range correlation does not predict performance. Judge calibration, a test-derived failure taxonomy, and a lucky-pass contamination bound were likewise post hoc and carry no multiplicity correction. Deviations, including a generation cap exceeded by $3.30 and a registered difficulty control replaced after it proved to be a near-oracle, are listed with the pre-registration in the released artifact.

a) Related work: Selecting the argmax of a learned proxy over-optimises it [7–9, 18]; statistically this is the optimizer's curse and structural selection bias [1, 19], and two recent notes draw the evaluation consequence: success counted before an answer is chosen is not deployed success [10], and best-of-n degrades calibration even under a perfect ranker [11]. Ensembles mitigate but do not eliminate it [20, 21], because capable models make correlated errors [2, 3], judge panels supply few effective votes [4, 5], common-mode error is

not identifiable from internal scores [6], and verification on code helps only on margin-pivotal cases [22]. Process and generative verifiers score and select [23, 24]. Our testbed is SWE-bench and its subsets with mini-SWE-agent [25–28]; execution-free SWE critics and trained failure locators target failure directly [12–14, 29], and agents declare success falsely and overconfidently [30], **but none conditions on an upstream selector, which is the gap this protocol fills.** Reward models and their standing [31–34]; on method we avoid DeLong for nested paired AUROC [35] and follow clustered ROC and registered-report practice [36–38].

V. Conclusion

A third to a half of what a reward-model critic confidently committed on two SWE benchmarks was broken. Across the configurations we pre-registered and ran, every interval for disagreement's incremental value included zero; the pooled estimates range from -0.030 to $+0.013$ and the loosest pooled 95% upper bound is $+0.12$ AUROC beyond the selector's own score. That bound remains above the $+0.05$ we declared actionable, so the result is a failure to detect a replicable gain, not equivalence or evidence that the gain is absent. One configuration crossed its bar on the pilot — maximum disagreement over pruned windows, 0.686 on twelve positives — and did not survive replication. The critics are stable and content-driven, and given the solution at length the best frontier judge ranks silent failures at 0.78–0.83; in these samples, their *disagreement* did not reveal the missing information. Anyone building an escalation trigger for critic-selected agents should validate it where it will be used — on the selected population, against the selector's own score, with executed gold and clustering on the real dependence unit. The stronger-selector cell is consistent with the same null but does not establish it.

Reproducibility. Code, pre-registrations with every dated amendment, every scored row, every critic's scores, the harness reports and a research log are in the released artifact. All figures regenerate from committed JSON.

References

[1] J. E. Smith and R. L. Winkler, "The optimizer's curse: Skepticism and postdecision surprise in decision analysis," *Management Science*, vol. 52, no. 3, pp. 311–322, 2006. [Online]. Available: https://doi.org/10.1287/mnsc.1050.0451

[2] S. Goel, J. Struber, I. A. Auzina *et al.*, "Great models think alike and this undermines ai oversight," *arXiv:2502.04313*, 2025. [Online]. Available: https://arxiv.org/abs/2502.04313

[3] E. Kim, A. Garg, K. Peng, and N. Garg, "Correlated errors in large language models," *arXiv:2506.07962*, 2025. [Online]. Available: https://arxiv.org/abs/2506.07962

[4] G. Kohli, "Nine judges, two effective votes: Correlated errors undermine llm evaluation panels," *arXiv:2605.29800*, 2026. [Online]. Available: https://arxiv.org/abs/2605.29800

[5] E. Hossain, N. Yousefi, and S.-N. Lim, "Agreement overstates evidence: Error dependence in llm judge consensus," *arXiv:2609.22512*, 2026. [Online]. Available: https://arxiv.org/abs/2609.22512

[6] F. Afrin and I. F. Shihab, "Evaluator ensembles under reward hacking: Covariance geometry and finite-search guarantees," *arXiv:2608.08002*, 2026. [Online]. Available: https://arxiv.org/abs/2608.08002

[7] L. Gao, J. Schulman, and J. Hilton, "Scaling laws for reward model overoptimization," *ICML*, 2023, arXiv:2210.10760. [Online]. Available: https://arxiv.org/abs/2210.10760

[8] H. Khalaf, C. Mayrink Verdun, A. Oesterling, H. Lakkaraju, and F. du Pin Calmon, "Inference-time reward hacking in large language models," *arXiv:2506.19248*, 2025. [Online]. Available: https://arxiv.org/abs/2506.19248

[9] Z. Yu, Z. S. Wu, and A. Block, "From curiosity to caution: Mitigating reward hacking for best-of-n with pessimism," in *ICLR*, 2026, arXiv:2604.04648. [Online]. Available: https://arxiv.org/abs/2604.04648

[10] S. M. Emadi, "When does a scaling result justify a different allocation? a critical review of resource-allocation evidence for ai systems," *arXiv:2609.14500*, 2026. [Online]. Available: https://arxiv.org/abs/2609.14500

[11] L. Lovén, N. Do, H. Mehmood, D. K. Sah, and S. Tarkoma, "The behavioral credibility trilemma: When calibrated autonomy becomes impossible," *arXiv:2605.25739*, 2026. [Online]. Available: https://arxiv.org/abs/2605.25739

[12] K. Shum, B. Hui, J. Chen, L. Zhang, J. Yang, Y. Huang, J. Lin, J. He *et al.*, "Swe-rm: Execution-free feedback for software engineering agents," *arXiv:2512.21919*, 2025. [Online]. Available: https://arxiv.org/abs/2512.21919

[13] S. Rahman, Y. Kim, M. Parmar *et al.*, "Locating hidden failures makes long-horizon agents more reliable," *arXiv:2609.17930*, 2026. [Online]. Available: https://arxiv.org/abs/2609.17930

[14] C. Wang, Y. Lyu, J. He, Z. Yang, C. Zhong, Y. Harel, and D. Lo, "Fail-fast, restart-smart: Early failure prediction and restart for swe agentic tasks," *arXiv:2608.03222*, 2026. [Online]. Available: https://arxiv.org/abs/2608.03222

[15] Y. Wang, M. Pradel, and Z. Liu, "Are "solved issues" in SWE-bench really solved correctly? an empirical study," *arXiv:2503.15223*, 2025. [Online]. Available: https://arxiv.org/abs/2503.15223

[16] S. Rajan, "Auditing reward hackability in code rl training environments," *arXiv:2606.16062*, 2026. [Online]. Available: https://arxiv.org/abs/2606.16062

[17] P. Sahoo, G. Mittal, X. Li, S. Ma, B. Steenhoek, P. Lin, and Y. Hu, "Agentlens: Revealing the lucky pass problem in swe-agent evaluation," *arXiv:2605.12925*, 2026.

[Online]. Available: https://arxiv.org/abs/2605.12925

[18] B. Stroebl, S. Kapoor, and A. Narayanan, "Inference scaling flaws: The limits of llm resampling with imperfect verifiers," *arXiv:2411.17501*, 2024. [Online]. Available: https://arxiv.org/abs/2411.17501

[19] M. A. Hernán, S. Hernández-Díaz, and J. M. Robins, "A structural approach to selection bias," *Epidemiology*, vol. 15, no. 5, pp. 615–625, 2004. [Online]. Available: https://doi.org/10.1097/01.ede.0000135174.63482.43

[20] J. Eisenstein, C. Nagpal, A. Agarwal, A. Beirami, A. D'Amour, D. Dvijotham, A. Fisch, K. Heller, S. Pfohl, D. Ramachandran, P. Shaw, and J. Berant, "Helping or herding? reward model ensembles mitigate but do not eliminate reward hacking," *arXiv:2312.09244*, 2023. [Online]. Available: https://arxiv.org/abs/2312.09244

[21] T. Coste, U. Anwar, R. Kirk, and D. Krueger, "Reward model ensembles help mitigate overoptimization," *ICLR*, 2024, arXiv:2310.02743. [Online]. Available: https://arxiv.org/abs/2310.02743

[22] Y. Shu, "Blind to the pivotal vote: Aggregate independence metrics miss where verification actually helps," *arXiv:2608.06940*, 2026. [Online]. Available: https://arxiv.org/abs/2608.06940

[23] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," *ICLR*, 2024, arXiv:2305.20050. [Online]. Available: https://arxiv.org/abs/2305.20050

[24] L. Zhang, A. Hosseini, H. Bansal, M. Kazemi, A. Kumar, and R. Agarwal, "Generative verifiers: Reward modeling as next-token prediction," *arXiv:2408.15240*, 2024. [Online]. Available: https://arxiv.org/abs/2408.15240

[25] C. E. Jimenez, J. Yang, A. Wettig, S. Yao, K. Pei, O. Press, and K. Narasimhan, "SWE-bench: Can language models resolve real-world github issues?" *ICLR*, 2024, arXiv:2310.06770. [Online]. Available: https://arxiv.org/abs/2310.06770

[26] OpenAI, "Introducing SWE-bench verified," https://openai.com/index/introducing-swe-bench-verified/, 2024.

[27] J. Yang, C. E. Jimenez *et al.*, "SWE-bench multilingual," *SWE-bench project, https://www.swebench.com/multilingual.html*, 2025. [Online]. Available: https://www.swebench.com/multilingual.html

[28] J. Yang, K. Lieret *et al.*, "mini-swe-agent," *https://github.com/SWE-agent/mini-swe-agent, version 2.4.6*, 2025. [Online]. Available: https://github.com/SWE-agent/mini-swe-agent

[29] A. Yadavally, H. Nguyen, L. Callot, and G. Guinet, "Large language model critics for execution-free evaluation of code changes," *arXiv:2501.16655*, 2025. [Online]. Available: https://arxiv.org/abs/2501.16655

[30] L. Advani, "From confident closing to silent failure: Characterizing false success in llm agents," *arXiv:2606.09863*, 2026. [Online]. Available: https://arxiv.org/abs/2606.09863

[31] C. Y. Liu, L. Zeng, Y. Xiao *et al.*, "Skywork-reward-v2: Scaling preference data curation via human-ai synergy," *arXiv:2507.01352*, 2025. [Online]. Available: https://arxiv.org/abs/2507.01352

[32] R. Yang, R. Ding, Y. Lin, H. Zhang, and T. Zhang, "Regularizing hidden states enables learning generalizable reward model for llms," *NeurIPS*, 2024, arXiv:2406.10216. [Online]. Available: https://arxiv.org/abs/2406.10216

[33] H. Zeng, D. Jiang, H. Wang, P. Nie, X. Chen, and W. Chen, "Acecoder: Acing coder rl via automated test-case synthesis," *arXiv:2502.01718*, 2025. [Online]. Available: https://arxiv.org/abs/2502.01718

[34] N. Lambert, V. Pyatkin, J. Morrison *et al.*, "Rewardbench: Evaluating reward models for language modeling," *arXiv:2403.13787*, 2024. [Online]. Available: https://arxiv.org/abs/2403.13787

[35] O. V. Demler, M. J. Pencina, and R. B. D'Agostino, "Misuse of delong test to compare aucs for nested models," *Statistics in Medicine*, vol. 31, no. 23, pp. 2577–2587, 2012. [Online]. Available: https://doi.org/10.1002/sim.5328

[36] N. A. Obuchowski, "Nonparametric analysis of clustered roc curve data," *Biometrics*, vol. 53, no. 2, pp. 567–578, 1997. [Online]. Available: https://doi.org/10.2307/2533958

[37] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *PNAS*, vol. 115, no. 11, pp. 2600–2606, 2018. [Online]. Available: https://doi.org/10.1073/pnas.1708274114

[38] C. D. Chambers and L. Tzavella, "The past, present and future of registered reports," *Nature Human Behaviour*, vol. 6, pp. 29–42, 2022. [Online]. Available: https://doi.org/10.1038/s41562-021-01193-7